

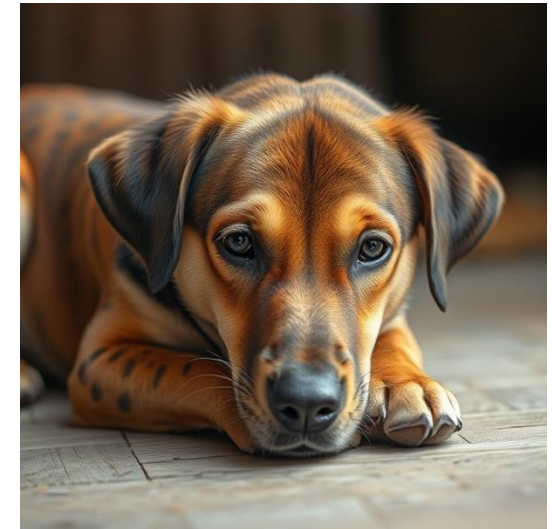


Machine learning for Data Engineers

Project Topics

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Deforestation, Climate Change, and Animal Welfare



Deforestation

A) Topics:

- **Predict Deforestation Rates:** Train a model to predict future deforestation areas based on features such as satellite imagery, land use, proximity to roads, and economic activities.
- **Forest Fire Risk Prediction:** Use historical weather data and vegetation data to predict the likelihood of forest fires in specific regions.
- **Species Habitat Prediction:** Predict areas where endangered species are most likely to thrive based on environmental features.

B) Possible datasets:

- [Global Forest Watch](#) - Deforestation and forest cover datasets.
- [MODIS Fire and Thermal Anomalies Data](#) - Satellite fire data.
- [WorldClim](#) - Global climate data for species modeling.
- [Copernicus Land Monitoring Service](#) - Provides land cover data including forests, urban areas, and agricultural land.
- [Inventaire Forestier National – IGN](#) - Data on forest types, density, and health across France.
- [Prométhée Database](#) - Contains forest fire records in France, including burned areas, causes, and dates.

Climate Change

A) Topics:

- **Carbon Footprint Estimation:** Predict the carbon emissions of industries or countries based on their economic and energy consumption patterns.
- **Climate Change Trends Analysis:** Use time-series analysis on temperature, CO2 concentration, and sea-level datasets to predict future climate change trends.
- **Renewable Energy Potential Prediction:** Build models to assess the potential of solar, wind, or hydroelectric power generation for specific regions.

B) Possible datasets:

- [Climate Data Store](#) - European climate data.
- [Carbon Dioxide Information Analysis Center](#) - CO2 emissions datasets.
- [Open Power System Data](#) - Renewable energy datasets.
- [RTE Open Data](#) - Data from France's electricity grid, including renewable energy generation (solar, wind, hydro) by region.
- [Météo France](#) - Historical and real-time weather data, including temperature, precipitation, and wind patterns.

Animal Welfare

A) Topics:

- **Animal Population Estimation:** Train a model to estimate populations of specific animals using drone imagery, thermal images, or audio recordings.
- **Animal Activity Recognition:** Classify animal behaviors (e.g., foraging, resting, moving) using sensor or video data.
- **Illegal Poaching Detection:** Detect poaching activities using spatial data and patterns of reported poaching incidents.

B) Possible datasets:

- [iNaturalist](#) - Biodiversity observations.
- [Global Biodiversity Information Facility \(GBIF\)](#) - Open biodiversity data.
- [Wildlife Insights](#) - Camera trap images for wildlife.
- [Vigie Nature](#) - Citizen science data on bird populations across France.

Multidisciplinary: Predicting Species Migration due to Climate Change

A) Topics:

- **Species Migration Patterns:** Predict how species' habitats will shift due to changes in temperature and precipitation using climate data and biodiversity records.
- **Impact of Urbanization on Biodiversity:** Analyze how urban expansion affects species diversity and suggest mitigation strategies.

B) Possible datasets:

- [CHELSA Climate Data](#) - High-resolution climate data for ecological modeling.
- [NASA Earth Data](#) - Climate and remote sensing data.
- [French Land Use Data \(BD TOPO\)](#) - Detailed maps of urban development across France.

Advanced Challenge: Predicting Renewable Energy Storage Needs

A) Topics:

- **Energy Demand and Storage Forecasting:** Predict renewable energy storage needs for a region based on energy generation patterns, weather forecasts, and consumption trends.
- **Battery Health Prediction:** Build a time-series model to predict the lifespan and efficiency of batteries used in renewable energy systems.

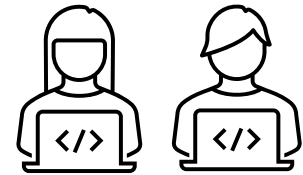
B) Possible datasets:

- [National Renewable Energy Laboratory \(NREL\)](#) - Renewable energy datasets.
- [Open Energy Platform](#) - Datasets on energy usage and generation.
- [Energy Information Administration \(EIA\)](#) - Global energy data.
- [Photovoltaic Geographical Information System \(PVGIS\)](#) - Solar and wind energy potential datasets.
- [IRENA \(International Renewable Energy Agency\)](#) - Reports and datasets on energy storage technologies in France and Europe.

Deliverables

Each group has 15 minutes maximum, to present the research presentation based on the outline shared “ProjectOutline.pdf”:

1. PPT to be uploaded in a shared folder on Teams
2. Code to be made public on GitHub (link provided in the PPT)



Tip: Start with simpler models (linear regression, random forests) and then build up to deep learning techniques